

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded such as $b(u) = u$. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

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1 Introduction

We consider the following *stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot). \end{cases} \quad (1.1) \quad \boxed{\text{E:SHE}}$$

The noise, $\dot{W}(t, s)$, is a centered Gaussian noise that is white in time and homogeneously colored in space defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with the natural filtration $\{\mathcal{F}_t\}_{t \geq 0}$ generated by the noise. Its covariance structure, J , is defined on as follows:

$$J(\psi, \phi) := \mathbb{E} [\dot{W}(\psi) \dot{W}(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad (1.2) \quad \boxed{\text{E:Cor}}$$

for all continuous and rapidly decreasing functions ψ and ϕ , where $\tilde{\phi}(x) := \phi(-x)$, “ $*$ ” refers to the convolution in spatial variable, and Γ is a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$ that is commonly referred to as the *correlation measure*. The Fourier transform of Γ (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure, which is usually called the *spectral measure* and is denoted by $\hat{f}(d\xi)$ ¹. Moreover, in the case where Γ has density f , namely, $\Gamma(dx) = f(x)dx$, then $\hat{f}(d\xi) = \hat{f}(\xi)d\xi$. The initial condition μ is a deterministic, locally finite, regular, signed Borel measure that satisfies the following integrability condition at the infinity (see (3.1) below).

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) < \infty \quad \text{for all } a > 0, \quad (1.3) \quad \boxed{\text{?E:Init?}}$$

where $|\mu| = \mu_+ + \mu_-$ and $\mu = \mu_+ - \mu_-$ refers to the *Hahn decomposition* of the measure μ . The function $b(x, u)$ is Lipschitz continuous in the second variable, uniformly in the first one, i.e., for some constants $L_b > 0$ and $L_0 \geq 0$,

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.4) \quad \boxed{\text{E:lipcon}}$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to the *parabolic Anderson model* [CM94]. The SPDEs (1.1) is understood in its *mild form*:

$$u(t, x) = J_0(t, x; \mu) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy), \quad (1.5) \quad \boxed{\text{E:gmsol}}$$

where $G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{1}{2t}|x|^2\right)$ is the heat kernel,

$$J_0(t, x) = J_0(t, x; \mu) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) \quad (1.6) \quad \boxed{\text{?E:Ju?}}$$

is the solution to the homogeneous equation, and the stochastic integral is the *Walsh integral*. We refer the interested readers to [Wal86; Dal99; Dal+09; CK19] for more details of the set up.

The aim of this paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system. The existence of invariant measure under the setting of the entire space $\mathbb{R}^d$ has been much less studied. In addition to Tessitore and Zabczyk [TZ98], with which we will follow closely in this paper, a few other papers that consider the whole space include [AM03; MSY20; MSY16]. Since the space $\mathbb{R}^d$ is not compact, one needs to work on a weighted space as in [TZ98]:

¹See Remark 2.10 for the convention of Fourier transform that is used in this paper.

$\langle \text{D:Rho} \rangle?$ **Definition 1.1.** $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an *admissible weight* if ρ is a strictly positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d. \quad (1.7) \quad \boxed{\text{E:aw}}$$

Let ρ be an admissible weight function, then we may introduce the Hilbert space $H := L^2_\rho(\mathbb{R}^d)$ with $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x) dx$ and $\|f\|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x) dx$. More discussions about the weight function can be found in Section 2.1. Let $\mathcal{B}(H)$ be the space of all bounded Borel functions on H . Following [DGZ92], a probability measure η on the Borel σ -field $\mathcal{B}(H)$ is said to be *invariant* for (1.1) if

$$\eta(A) = \int_H \mathcal{L}(u(t, \cdot; \zeta))(A) \eta(d\zeta), \quad \text{for all } t \geq 0 \text{ and } A \in \mathcal{B}(H), \quad (1.8) \quad \boxed{\text{E:InvMeas?}}$$

where we use the notation $u(t, x; \zeta)$ with a third argument to emphasize its dependence on the initial condition $u(0, \cdot) = \zeta$ and $\mathcal{L}(u(t, \cdot; \zeta))(A)$ denotes the law of the solution:

$$\mathcal{L}(u(t, \cdot; \zeta))(A) = \mathbb{P}[\omega \in \Omega : u(t, \cdot; \zeta)(\omega) \in A], \quad A \in \mathcal{B}(H).$$

Due the *Krylov-Bogoliubov theorem* (see, e.g., [DZ14, Theorem 11.7]), once the tightness of $\{\mathcal{L}(u(t, \cdot; \mu))\}_{t \geq t_0}$ for some $t_0 \geq 0$ is established, one can construct the invariant measure via

$$\eta(A) = \lim_{n \rightarrow \infty} \frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u(t, \cdot; \mu))(A) dt, \quad (1.9) \quad \boxed{\text{E:InvMeasForm}}$$

for some sequence $\{T_n\}_{n \geq 1}$ with $T_n \uparrow \infty$. A critical step in obtaining tightness is to show that the following moment is uniformly bounded in time:

$$\sup_{t \geq 0} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) < \infty. \quad (1.10) \quad \boxed{\text{E:BddMnt}}$$

Note that the uniqueness of the invariant measure is a much harder question; see, e.g., [HM06].

It is known that the solution to (1.1) is usually *intermittent*, namely, the probability moments to (1.1) have a certain exponential growth in t ; see, e.g., [CM94; FK09]. In [AM03; MSY20; MSY16], a drift (or reaction-diffusion) term $f(x, u)$ is included in (1.1) in a crucial way to help cancel the otherwise exponential growing moments. The absence of such a drift term in this paper makes the problem more challenging. In order to have moments bounded in time, as required in (1.10), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [CK19, Theorem 1.3 and Lemma 2.5], where necessary and sufficient conditions are given. More precisely, for bounded second moments one has to have the spatial dimension $d \geq 3$, and in addition, the spectral measure, $\hat{f}$, and Lipschitz constant, L_b , (the Lyapunov exponent of b) in (1.4) have to satisfy the following two conditions:

$\langle \text{E:Uofinite} \rangle$

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{|\xi|^2} < \infty \quad (1.11a) \quad \boxed{\text{E:Dalang-00}}$$

$$\text{and } 64L_b^2 < \frac{1}{4\Upsilon(0)}. \quad (1.11b) \quad \boxed{\text{E:Lipb}}$$

Note that condition (1.11a) is a strengthened version of *Dalang's condition*:

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty, \quad \text{for some (and hence) all } \beta > 0. \quad (1.12) \quad \boxed{\text{E:Dalang}}$$

Recall that in order to obtain the Hölder continuity of the solution, one needs to strengthen (1.12) in a different way as follows, for some $\alpha \in (0, 1]$,

$$\Upsilon_\alpha(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(\mathrm{d}\xi)}{(\beta + |\xi|^2)^{1-\alpha}} < \infty \quad \text{for some (hence all) } \beta > 0; \quad (1.13) \quad \boxed{\text{E:Dalang-ab}}$$

see Theorem 1.8 of [CH19] or [SS02]. Similarly, one can further strengthen condition (1.13) to

$$\Upsilon_\alpha(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(\mathrm{d}\xi)}{|\xi|^{2(1-\alpha)}} < \infty \quad \text{for some } \alpha \in (0, 1]. \quad (1.14) \quad \boxed{\text{E:Dalang-a0}}$$

We use the convention that when $\alpha = 0$, we simply drop it from the expression $\Upsilon_\alpha(\beta)$, i.e., $\Upsilon(\beta) = \Upsilon_0(\beta)$. The relations of these conditions are shown in the Figure 1.

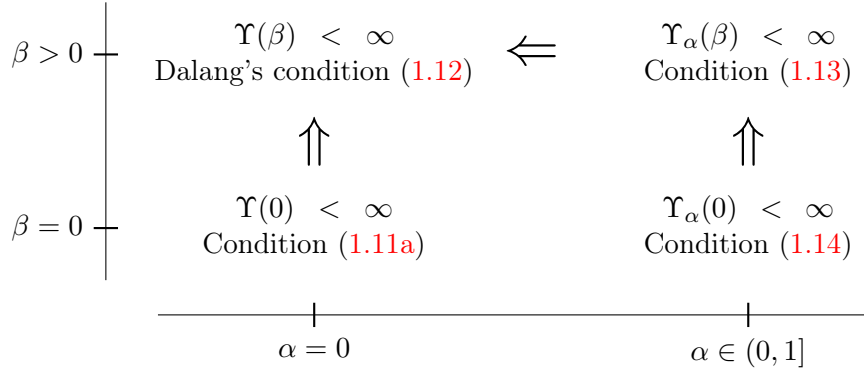


Figure 1: Relations among conditions (1.12), (1.13), (1.14) and (1.11a). Check also Lemma 4.5 for the relation between $\Upsilon(0) < \infty$ and $\Upsilon_\alpha(0) < \infty$.

(F:Dalang)

Note that the two conditions in (1.11a) guarantee the existence of the following non-empty open interval:

$$(2^8 L_b^2 \Upsilon(0), 1) \neq \emptyset. \quad (1.15) \quad \boxed{\text{E:Interval}}$$

Now we are ready to state our two main results of the paper:

(T:Lrho) **Theorem 1.2.** *Let $u(t, x; \mu)$ be the solution to (1.1) starting from μ . Assume that*

- (i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is an nonnegative $L^1(\mathbb{R}^d)$ function;
- (ii) the initial condition μ satisfies

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} J_0^2(t, x; 1 + |\mu|) \rho(x) \, dx < \infty \quad \text{for all } t > 0; \quad (1.16) \quad \boxed{\text{E:G}}$$

- (iii) the spectral measure $\widehat{f}$ and the Lipschitz constant L_b satisfy the two conditions in (1.11).

Then there exists an unique $L^2(\Omega)$ -continuous solution $u(t, x)$ such that for some constant $C > 0$, which does not depend on t , and for all $t > 0$ the following holds:

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu) < \infty. \quad (1.17) \quad \boxed{\text{E:uInLrho}}$$

This theorem will be proved in Section 3.

Theorem 1.3. *Let $u(t, x)$ be the solution to (1.1) starting from μ and let ρ be an admissible weight function. Assume that*

(i) *there exists another admissible weight $\tilde{\rho}$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\tilde{\rho}(x)} dx < \infty; \quad (1.18) \quad \text{E:rho}$$

(ii) *the weight function $\tilde{\rho}$ and the initial condition satisfy the following condition:*

$$\limsup_{t>0} \mathcal{G}_{\tilde{\rho}}(t; \mu) < \infty; \quad (1.19) \quad \text{E:InitData}$$

(iii) *the spectral measure $\hat{f}$ and the Lipschitz constant L_b satisfy the two conditions in (1.11);*

(iv) *for some $\alpha \in (2^8 \Upsilon(0) L_b^2, 1)$ (see (1.15)), the spectral measure $\hat{f}$ satisfies (1.14).*

Then we have that

(a) *for any $\tau > 0$, the sequence of laws of $\{\mathcal{L}u(t, \cdot; \mu)\}_{t \geq \tau}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L_\rho^2(\mathbb{R}^d)$ such that*

$$\mathcal{L}u(t, \cdot; \mu)(\mathcal{K}) := \mathbb{P}(u(t, \cdot; \mu) \in \mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau > 0; \quad (1.20) \quad \text{E:cpt}$$

(b) *there exists an invariant measure for the laws $\{\mathcal{L}u(t, \cdot; \mu)\}_{t>0}$ in $L_\rho^2(\mathbb{R}^d)$.*

This theorem will be proved in Section 5.

The work by Tessitore and Zabczyk [TZ98] has a strong influence in the current work. We formulate and solve the problem using the random field language with some state-of-art moment estimates. The improvements over [TZ98] consist of the following aspects:

1. Results in [TZ98] allows essentially all bounded functions as the initial conditions, though Theorem 3.3 (*ibid.*) was proved only for the constant one initial condition. Here we give precise condition on the initial condition, namely, (1.19), which allows a much wider class of initial conditions, including unbounded functions and measures such as the Dirac delta measure; see Examples 2.4 and 2.5. We emphasize that the Dirac delta initial measure plays a very prominent role in the study of the stochastic heat equation; see, e.g., [ACQ11].
2. We give ...

This paper is organized as follows: In Section 2, we will further discuss our main results and provide some examples. In particular, in Section 2.1 we make some comments on the weight function and in Section 2.2 we show that our results could include a wider class of initial conditions. Finally, the two main Theorems 1.2 and 1.3 will be proved in Sections 3 and 5, respectively.

Throughout the paper, we will use $\|X\|_p$ to denote the $L^p(\Omega)$ norm, namely, $(\mathbb{E}(|X|^p))^{1/p}$. We will use several times the following factorization property of the heat kernel

$$G(t, x)G(s, y) = G\left(\frac{ts}{t+s}, \frac{sx+ty}{t+s}\right) G(t+s, x-y), \quad (1.21) \quad \text{E:gsim}$$

which can be easily verified. We also will repeatedly use the following spherical coordinate formula:

$$\int_{\mathbb{R}^d} f(|x|) dx = \sigma(\mathbb{S}^{d-1}) \int_0^\infty f(r) r^{d-1} dr,$$

and we may from time to time use the following $\sigma(\mathbb{S}^{d-1}) = 2(\pi)^{d/2}/\Gamma(d/2)$. The convention of Fourier transform is given in Remark 2.10 below.

2 Discussions and Examples

$\langle \text{S:Example} \rangle$ 2.1 Weight functions

$\langle \text{SS:Weight} \rangle$ Here are several examples of admissible weights (see Example in Section 2 of [TZ98])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \tag{2.1} \quad \boxed{\text{E:admRho}}$$

$\langle \text{R:admRho} \rangle$ **Remark 2.1.** The smaller the weight function $\rho(\cdot)$ (not necessarily admissible) is, the larger the space $L_\rho^2(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (2.1) (up to a constant). However, one can easily check that the admissible condition (1.7) excludes these two cases. It is much less obvious that the following weight function is not admissible:

$$\rho(x) = \exp\left(-a|x|^b\right), \quad x \in \mathbb{R}^d, \quad \text{with } a > 0 \text{ and } b \in (1, 2) \text{ fixed.}$$

The admissible condition 1.7 is needed in this paper due to the following result:

$\langle \text{P:weight} \rangle$ **Proposition 2.2** (Proposition 2.1 of [TZ98]). *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L_ρ^2 . Moreover, if $\tilde{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\tilde{\rho}(x)} dx < \infty,$$

then for any $t > 0$, the operators defined above are compact from $L_\rho^2(\mathbb{R}^d)$ to $L_{\tilde{\rho}}^2(\mathbb{R}^d)$.

2.2 Various initial conditions

$\langle \text{Ex:Riesz} \rangle$ **Example 2.3** ($L^\infty(\mathbb{R}^d)$ initial condition). We emphasize that if the initial condition μ is deterministic and is such that $\mu(dx) = \varphi(x)dx$ with $\varphi \in L^\infty(\mathbb{R}^d)$, then all conditions related to $\mathcal{G}_\rho(\cdot)$ in both Theorems 1.2 and 1.3 are trivially satisfied. To be more precise, both conditions (1.16) and (1.19) hold because

$$\mathcal{G}_\rho(t; \varphi) \leq \|\varphi\|_{L^\infty(\mathbb{R}^d)}^2 \|\rho\|_{L^1(\mathbb{R}^d)} < \infty \quad \text{uniformly for all } t \geq 0.$$

$\langle \text{Ex:delta} \rangle$ **Example 2.4** (Delta initial condition). In this example, we study the case when the initial condition μ is the Dirac delta measure at zero, namely, δ_0 . Let ρ be a nonnegative $L^1(\mathbb{R}^d)$ function. Since

$$\mathcal{G}_\rho(t; \delta_0) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx \leq G(t, 0)^2 \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0,$$

we see that both conditions (1.16) and (1.19) are satisfied. In particular, $\limsup_{t \rightarrow 0} \mathcal{G}_\rho(t; \delta_0) = 0$.

$\langle \text{Ex:Riesz} \rangle$ **Example 2.5** (More initial conditions not in $L_\rho^2(\mathbb{R}^d)$). In this example, we study the case when $\mu(dx) = |x|^{-\alpha} dx$ for some $\alpha \in (0, d)$. It is clear that when $\alpha \in (d/2, d)$, $\mu \notin L_\rho^2(\mathbb{R}^d)$. However, in this case, we have

$$J_0(t, x) = (G(t, \cdot) * |\cdot|^{-\alpha})(x) \leq (G(t, \cdot) * |\cdot|^{-\alpha})(0).$$

On the other hand, Hence,

$$(G(t, \cdot) * |\cdot|^{-\alpha})(0) = \frac{2\pi^{d/2}}{\Gamma(d/2)} \times (2\pi t)^{-d/2} \int_0^\infty e^{-\frac{r^2}{2t}} r^{-\alpha+d-1} dr = C_* t^{-\alpha/2},$$

with $C_* = 2^{-\alpha/2} \Gamma((d - \alpha)/2) / \Gamma(d/2)$, which implies that

$$\mathcal{G}_\rho(t; |\cdot|^{-\alpha}) \leq \int_{\mathbb{R}^d} J_0^2(t, 0) \rho(x) dx = C_*^2 t^{-\alpha} \|\rho\|_{L^1(\mathbb{R}^d)}, \quad \text{for all } t > 0.$$

Therefore, we see that both conditions (1.16) and (1.19) are satisfied.

The following proposition shows that for initial conditions with unbounded tails, condition (1.16) may hold while condition (1.19) may fail.

Proposition 2.6. *Suppose that $\rho = \exp(-|x|)$, which is an admissible weight function. Let the initial condition μ be given as*

$$\mu(dx) = |x|^\alpha dx \quad \text{with } \alpha > 0.$$

Then for some constants $C, C' > 0$ that depend on d and α , it holds that

$$C'(1 + t^\alpha) \leq \mathcal{G}_\rho(t; \mu) \leq C(1 + t^\alpha), \quad \text{for all } t > 0. \quad (2.2) \quad \boxed{\text{E:growtail}}$$

In particular, this implies that condition (1.16) is satisfied, but condition (1.19) fails.

Proof. Notice that by scaling arguments,

$$\begin{aligned} \mathcal{G}_\rho(t; \mu) &= \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y) |y|^\alpha dy \right)^2 e^{-|x|} dx \\ &= \int_{\mathbb{R}^d} t^\alpha \left(\int_{\mathbb{R}^d} G\left(1, \frac{x}{\sqrt{t}} - z\right) |z|^\alpha dz \right)^2 e^{-|x|} dx \\ &= \int_{\mathbb{R}^d} t^{\alpha+d/2} \left(\int_{\mathbb{R}^d} G(1, \xi - z) |z|^\alpha dz \right)^2 e^{-\sqrt{t}|\xi|} d\xi. \end{aligned}$$

In the following, let $C, C_\alpha, C'_\alpha, C_{\alpha,d}$ and $C'_{\alpha,d}$ be generic constants that may depend on α and d and may change their value at each appearance.

Upper bound: Because

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \leq C_\alpha \int_{\mathbb{R}^d} G(1, z) (|\xi|^\alpha + |z|^\alpha) dz = C'_\alpha (1 + |\xi|^\alpha),$$

I believe for the last equality to hold, this would force C'_α to also depend on ξ . This issue is resolved if we just replace $=$ with $\leq$ and then let $C'_\alpha = \max\{1, \int_{\mathbb{R}^d} G(1, z) |z|^\alpha dz\}$

we see that

$$\begin{aligned} \mathcal{G}_\rho(t; \mu) &\leq C_\alpha \int_{\mathbb{R}^d} t^{\alpha+d/2} (1 + |\xi|^{2\alpha}) e^{-\sqrt{t}|\xi|} d\xi \\ &= C_{\alpha,d} \int_0^\infty t^{\alpha+d/2} (1 + r^{2\alpha}) e^{-\sqrt{t}r} r^{d-1} dr \\ &= C_{\alpha,d} (t^\alpha \Gamma(d) + \Gamma(d + 2\alpha)) \\ &= C'_{\alpha,d} (1 + t^\alpha) < \infty. \end{aligned}$$

Hence, this proves the upper bound in (2.2).

Lower bound: Now we prove the lower bound in (2.2). Indeed,

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \geq \int_{\mathbb{R}^d} G(1, z) ||\xi| - |z||^\alpha dz$$

$$\begin{aligned}
&\geq C_d \int_0^\infty ||\xi| - x|^\alpha e^{-\frac{x^2}{2}} x^{d-1} dx \\
&\geq C_d \int_1^2 ||\xi| - x|^\alpha dx \\
&= \frac{C_d}{1+\alpha} \psi(|\xi|),
\end{aligned}$$

where, by considering three cases, we have

$$\begin{aligned}
\psi(r) &= \begin{cases} (2-r)^{\alpha+1} - (1-r)^{\alpha+1} & \text{if } 0 < r < 1, \\ (2-r)^{\alpha+1} + (r-1)^{\alpha+1} & \text{if } 1 \leq r \leq 2, \\ (r-1)^{\alpha+1} - (r-2)^{\alpha+1} & \text{if } r > 2, \end{cases} \\
&= \operatorname{sgn}(2-r)|r-2|^{\alpha+1} + \operatorname{sgn}(r-1)|r-1|^{\alpha+1}.
\end{aligned}$$

We claim that

$$\inf_{r \geq 0} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0. \quad (2.3) \quad \boxed{\text{E:psi}}$$

With (2.3), we have that

$$\int_{\mathbb{R}^d} G(1, z) |\xi - z|^\alpha dz \geq C_{\alpha, d} \sqrt{1 + |\xi|^{2\alpha}}.$$

Then, by the same arguments as above for the upper bound, we obtain the lower bound in (2.2). It remains to prove (2.3), which will be proved in three cases.

When $r > 2$, we see that

$$\begin{aligned}
\frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} &\geq C_\alpha \frac{(r-1)^{\alpha+1} - (r-2)^{\alpha+1}}{(1+r)^\alpha} \\
&\geq C_\alpha \frac{(r-1)^\alpha(r-1) - (r-1)^\alpha(r-2)}{(1+r)^\alpha} \\
&= C_\alpha \left(\frac{r-1}{1+r} \right)^\alpha = C_\alpha \left(1 - \frac{2}{1+r} \right)^\alpha \geq C_\alpha \left(1 - \frac{2}{3} \right)^\alpha,
\end{aligned}$$

where in the first inequality, we considered two cases: $2\alpha \geq 1$ and $2\alpha < 1$. When $2\alpha < 1$, we used concavity to say that $(1+r^{2\alpha})^{2\alpha}/2 \leq ((1+r)/2)^{2\alpha}$. When $2\alpha \geq 1$ we used the fact that for $a, b > 0$ that $a^{2\alpha} + b^{2\alpha} \leq (a+b)^{2\alpha}$. This shows that $\inf_{r > 2} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} > 0$.

When $r \in (1, 2]$, elementary calculations show that the minimum of $\psi(r)$ is achieved at $r = 3/2$. Hence,

$$\inf_{r \in (1, 2]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(3/2)}{\sqrt{1+4^\alpha}} > 0.$$

Similarly, when $r \in (0, 1]$, by differentiation, one finds that the function $\psi(r)$ is nonincreasing. Hence, the minimum is achieved at $r = 1$:

$$\inf_{r \in (0, 1]} \frac{\psi(r)}{\sqrt{1+r^{2\alpha}}} \geq \frac{\psi(1)}{\sqrt{2}} > 0.$$

Combing the above three cases proves (2.3). This completes the proof of Proposition 2.6. $\square$

2.3 Bessel kernel and Matérn class of correlation functions

Example 2.7 (Bessel kernel as a correlation function). Let f_s denote the Bessel kernel with a strictly positive parameter $s > 0$. It is known that (see, e.g., Section 1.2.2 of [Gra14])

1. $f_s(x) > 0$ for all $x \in \mathbb{R}^d$ and $\|f_s\|_{L^1(\mathbb{R}^d)} = 1$;
2. there exists a constant $C(s, d) > 0$ such that

$$f_s(x) \leq C(s, d) \exp(-|x|/2) \quad \text{for } |x| \geq 2;$$

3. there exists a constant $c(s, d) > 0$ such that

$$\frac{1}{c(s, d)} \leq \frac{f_s(x)}{H_s(x)} \leq c(s, d) \quad \text{for } |x| \leq 2, \quad \text{with}$$

$$H_s(x) = \begin{cases} |x|^{s-d} + 1 + O(|x|^{s-d+2}) & \text{for } 0 < s < d, \\ \log\left(\frac{2}{|x|}\right) + 1 + O(|x|^2) & \text{for } s = d, \\ 1 + O(|x|^{s-d}) & \text{for } s > d; \end{cases}$$

4. the Fourier transform of f_s is strictly positive:

$$\mathcal{F}f_s(\xi) = \frac{1}{(1 + |\xi|^2)^{s/2}}. \quad (2.4) \quad \boxed{\text{E:FB}}$$

Indeed, one can use (2.4) as the definition of the Bessel kernel. Properties 1 and 4 ensure that f_s is a nonnegative and nonnegative-definite tempered measure for all $s > 0$. From Property 4, we see that

$$\Upsilon(0) < \infty \quad \Longleftrightarrow \quad s > d - 2.$$

In the following, we will assume that $s > d - 2$ and $d \geq 3$.

Example 2.8 (Matérn class of correlation functions). The *Matérn class of correlation functions* have been widely used in spatial statistics; one may check a recent work [LSW21] for references. Following Section 2.10 of [Ste99], this class of correlation functions are given by

$$K(x) = \phi \cdot (\alpha|x|)^\nu \mathcal{K}_\nu(\alpha|x|), \quad \text{for } x \in \mathbb{R}^d \text{ with } \phi > 0, \alpha > 0, \nu > 0, \quad (2.5) \quad \boxed{\text{E:Matern}}$$

where $\mathcal{K}_\nu(\cdot)$ is the modified Bessel function of second type, α and ν refer to the *scaling and smoothness parameters*, respectively. From the inversion formula (see p. 46 *ibid.*), one sees that

$$\mathcal{F}K(\xi) = (2\pi)^d \mathcal{F}^{-1}K(\xi) = (2\pi)^d f(|\xi|) \quad \text{with} \quad f(\xi) = \frac{2^{\nu-1} \phi \Gamma(\nu + d/2) \alpha^{2\nu}}{\pi^{d/2} (\alpha^2 + |\xi|^2)^{\nu+d/2}}, \quad \xi \in \mathbb{R}^d.$$

Comparing the above expression with (2.4), we see that the class of Bessel kernel f_s , with $s > d - 2$ and $d \geq 3$, includes the Matérn class (2.5) as a special case under the following choice of parameters:

$$\alpha = 1, \quad \nu = (s - d)/2, \quad \text{and} \quad \phi = 2^{(2-d-s)/2} \pi^{-d/2} \Gamma(s/2)^{-1}.$$

Note that the requirement of the smoothness parameter $\nu > 0$ for the Matérn class corresponds to the case of the Bessel kernel with $s > d$.

(P:Bessel)

Proposition 2.9. For the Bessel kernel $f_s(\cdot)$ with $s > 0$ defined in Example 2.7, it holds that

$$\Upsilon(0) = \frac{2\Gamma((s-d+2)/2)}{\pi^{d/2}(d-2)\Gamma(s/2)}, \quad \text{for all } s > d-2 \text{ and } d > 2, \text{ and} \quad (2.6) \quad \boxed{\text{E:UpsionB}}$$

$$\mathcal{H}_\alpha(t) < \infty \quad \forall t > 0 \quad \iff \quad 0 < \alpha < \frac{1}{2} - \frac{(d-s)_+}{4} \quad \text{and} \quad s > d-2 > 0, \quad (2.7) \quad \boxed{\text{E:HaB}}$$

where $a_+ := \max(a, 0)$ and $\mathcal{H}_\alpha(t)$ is defined in (4.2) below. Moreover, for $\alpha \in (0, 1/2)$, we have the following series expansion of $\mathcal{H}_\alpha(t)$ at $t \rightarrow 0$:

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2}\Gamma((d-s)/2)}{((s-d)/2+1-2\alpha)\Gamma(d/2)} t^{(s-d)/2+1-2\alpha} \\ \quad - \frac{\pi^{d/2}\Gamma((s-d)/2)}{(1-2\alpha)\Gamma(s/2)} t^{1-2\alpha} & d-2 < s < d \\ \quad + O\left(t^{(s-d)/2+2(1-\alpha)}\right), & \text{and} \\ & \alpha < \frac{1}{2} - \frac{1}{4}(d-s) \end{cases} \quad (2.8-a) \quad \boxed{\text{E:HaBS}}$$

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2}}{(1-2\alpha)\Gamma(d/2)} t^{1-2\alpha} \log\left(\frac{1}{t}\right) \\ \quad + \frac{\pi^{d/2}(1-(1-2\alpha)[\psi(d/2)+2\gamma])}{(1-2\alpha)^2\Gamma(d/2)} t^{1-2\alpha} & s = d \\ \quad + O(t^2 \log(t)), \end{cases} \quad (2.8-b) \quad \boxed{\text{E:HaBS}}$$

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2}\Gamma((s-d)/2)}{(1-2\alpha)\Gamma(s/2)} t^{1-2\alpha} + O(t^{(s-d)/2+1-2\alpha}), & d < s < d+2 \end{cases} \quad (2.8-c)$$

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2}}{(1-2\alpha)\Gamma(d/2+1)} t^{1-2\alpha} + O(t^{2(1-\alpha)} \log(t)), & s = d+2 \end{cases} \quad (2.8-d)$$

$$\mathcal{H}_\alpha(t) = \begin{cases} \frac{\pi^{d/2}}{(1-2\alpha)\Gamma(d/2+1)} t^{1-2\alpha} + O(t^{2(1-\alpha)}), & s > d+2 \end{cases} \quad (2.8-e)$$

where $\psi(x) = \frac{d}{dx} \log \Gamma(x)$ refers to the digamma function and $\gamma \approx 0.57721$ to Euler's constant; see, e.g., 5.2.2 and 5.2.3 on p. 136 of [AR10].

Proof. By the spherical coordinates and (2.4), for all $\alpha \in [0, 1)$,

$$\Upsilon_\alpha(0) = (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{d\xi}{|\xi|^{2(1-\alpha)} (1+|\xi|^2)^{s/2}} = (2\pi)^{-d} C_d \int_0^\infty \frac{r^{d-1}}{r^{2(1-\alpha)} (1+r^2)^{s/2}} dr,$$

where $C_d := \frac{2\pi^{d/2}}{\Gamma(d/2)}$. Now by change of variable $z = r^2/(1+r^2)$, we can evaluate the above integral explicitly by transforming it to the Beta integral:

$$\begin{aligned} \int_0^\infty \frac{r^{d-1}}{r^{2(1-\alpha)} (1+r^2)^{s/2}} dr &= \frac{1}{2} \int_0^1 z^{d/2+\alpha-2} (1-z)^{(s-d)/2-\alpha} dz \\ &= \frac{\Gamma(d/2-1+\alpha) \Gamma((s-d)/2+1-\alpha)}{2\Gamma(s/2)}, \end{aligned}$$

which is finite provided that $s > d - 2(1 - \alpha) > 0$. This proves (2.6).

It remains to prove (2.8), which then implies (2.7). From (4.2) and by the spherical coordinates, for all $t > 0$,

$$\begin{aligned}\mathcal{H}_\alpha(t) &= \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} d\xi \frac{\exp(-r|\xi|^2)}{(1 + |\xi|^2)^{s/2}} \\ &= C_d \int_0^t dr r^{-2\alpha} \int_0^\infty dz \frac{\exp(-rz^2)}{(1 + z^2)^{s/2}} z^{d-1} \\ &= \frac{C_d}{2} \int_0^t dr r^{-2\alpha} \int_0^\infty du \exp(-ru) (1 + u)^{-s/2} u^{d/2-1} \\ &=: \frac{C_d \Gamma(d/2)}{2} \int_0^t dr r^{-2\alpha} I(r) = \pi^{d/2} \int_0^t dr r^{-2\alpha} I(r).\end{aligned}$$

By [Old10, 13.4.4 on p.326], $I(r)$ is equal to the *confluent hypergeometric function*:

$$I(r) = U\left(\frac{d}{2}, \frac{2 + d - s}{2}, r\right).$$

By 18.2.18 – 13.2.22 on p. 323 *ibid.*, we see that

$$I(r) = \begin{cases} \frac{\Gamma((d-s)/2)}{\Gamma(d/2)} r^{(s-d)/2} + \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2+1}) & d-2 < s < d & 18.2.18, \\ -\frac{1}{\Gamma(d/2)} (\log(r) + \psi(d/2) + 2\gamma) + O(r \log(r)) & s = d & 18.2.19, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r^{(s-d)/2}) & d < s < d+2 & 18.2.20, \\ \frac{1}{\Gamma(d/2+1)} + O(r \log(r)) & s = d+2 & 18.2.21, \\ \frac{\Gamma((s-d)/2)}{\Gamma(s/2)} + O(r) & s > d+2 & 18.2.22. \end{cases}$$

Then integrating the right-hand side of the above expressions against $\pi^{d/2} r^{-2\alpha} dr$ over $[0, t]$ gives the five cases in (2.8). This completes the proof of Proposition 2.9. $\square$

2.4 Comparing Tessitore and Zabczyk's condition

?(SS:TessZabz)? In the setting of equation (1.1), Tessitore and Zabczyk [TZ98] prove existence of an invariant measure for (1.1) in $L_\rho^2(\mathbb{R}^d)$ under the assumption that there exists a $\varphi \in L_\rho^2(\mathbb{R}^d) \cap L_{\tilde{\rho}}^2(\mathbb{R}^d)$ where $\rho(\tilde{\rho})^{-1} \in L^1(\mathbb{R}^d)$ and the solution starting from φ is bounded in probability in $L_{\tilde{\rho}}^2(\mathbb{R}^d)$ and also that the spectral density $\hat{f}$ satisfies

$$\hat{f} \in L^p(\mathbb{R}^d) \quad \text{where} \quad \frac{d-2}{d} < \frac{1}{p}; \quad (2.9) \quad \text{?E:CorTZ?}$$

see Hypothesis 2.1 (*ibid.*). However as was illustrated in [TZ98, Theorem 3.3], in order to apply this theorem (or to have moments uniformly bounded in time), the following additional assumptions were imposed:

$$d \geq 3 \quad \text{and} \quad L_b^{-2} > \frac{\Gamma(d/2-1)2^{d/2-2}}{(2\pi)^{2d}} \int_{\mathbb{R}^d} \left(\left| \mathcal{F}\left(\sqrt{\hat{f}}\right) \right| * \left| \mathcal{F}\left(\sqrt{\hat{f}}\right) \right| \right) (\zeta) |\zeta|^{2-d} d\zeta, \quad (2.10) \quad \boxed{\text{E:TZspec}}$$

where Γ refers to the Gamma function; See Remark 2.10. With these assumptions, they were able to prove that (1.1) starting from the constant 1 satisfies (1.10) and thus is bounded in probability, verifying the existence of an invariant measure. Moreover, the measure takes the form of (1.9) above with $\varphi = 1$.

The additional assumptions given by Tessitore and Zabczyk parallel our assumptions (1.11a) and (??). However, it is not at all trivial to check which spectral densities, $\hat{f}(d\xi)$, satisfy (2.10) and no such examples were given in [TZ98]. For example, it may not be obvious from (2.10) that one can not take the spectral density of the noise to be the Riesz kernel, $\hat{f}(\xi) = |\xi|^{d-\alpha}$, for any $0 < \alpha < d$ but this can be easily seen from our condition (1.11a). The condition that $d \geq 3$ forces us to only consider a colored noise as in (1.2) and so the loss of the ability to apply the Riesz kernel is unfortunate. However, as we will see in Example 2.7 below, we are able to apply the Bessel kernel, $B_s(x)$, with $s < d$, which has exponential decaying tails and power growth at the origin. So in absence of the Riesz kernel, we still may apply our results to a kernel, namely the Bessel kernel, that has origin blow up.

Moreover by applying a restart argument (see Remark ?? and Example 2.7 below), our method allows us to verify (1.10), and in return (1.9), for more general initial conditions $\mu(\cdot)$ which include:

$$\text{Dirac delta measure } \delta_0(dx) \quad \text{and} \quad \varphi(x)dx \text{ with } \varphi \in C^\infty(\mathbb{R}^d).$$

Thus, we can expand our choice of initial data to include singular distributions, which are outside the space of $L^2_\rho(\mathbb{R}^d)$. We will expand on this in the following remark.

(R:Fourier) Remark 2.10. The Fourier transform may be defined differently depending how to handle the 2π constant. In this paper (as in [CK19; CH19]), we use the convention that

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) := \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi.$$

Hence, the Plancherel theorem takes the form of

$$\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\psi}(\xi) \overline{\widehat{\phi}(\xi)} d\xi. \quad (2.11) \quad \boxed{\text{E:Plancherel}}$$

The authors did not explicitly mention in [TZ98] about the convention of the Fourier transform. However, the proof of Theorem 3.3 (*ibid.*) suggests that the following convention has been used

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi.$$

Hence, the Plancherel theorem takes the form of $\int_{\mathbb{R}^d} \psi(x) \phi(x) dx = \int_{\mathbb{R}^d} \widehat{\psi}(\xi) \overline{\widehat{\phi}(\xi)} d\xi$ without the additional factor $(2\pi)^{-d}$. When we translate their condition (3.4) (*ibid.*) to (2.10), we have taken into account of this difference. In particular, the spectral density γ (*ibid.*) corresponds to $(2\pi)^{-d/2} \widehat{f}$ in this paper. This accounts the slightly different factor in front of the integral in (2.10) from that in (3.4) (*ibid.*).

Let us denote the right-side of the second condition in (2.10) by $M(f)$, namely,

$$M(f) := \Gamma(d/2 - 1) 2^{d/2-2} \int_{\mathbb{R}^d} \left(\left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| * \left| \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right| \right) (\zeta) |\zeta|^{2-d} d\zeta,$$

and consider two cases:

Case I: If $\widehat{f}(\zeta)$ is strictly positive for all $\zeta \in \mathbb{R}^d$, then there is no ambiguity when taking square root and hence one can remove the absolute value to see that

$$\begin{aligned} M(f) &= \Gamma(d/2 - 1) 2^{d/2-2} \int_{\mathbb{R}^d} \left(\mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) * \mathcal{F}^{-1} \left(\sqrt{\widehat{f}} \right) \right) (\zeta) |\zeta|^{2-d} d\zeta \\ &= \frac{1}{(2\pi)^{3d/2}} \int_{\mathbb{R}^d} \widehat{f}(\zeta) |\zeta|^{-2} d\zeta = (2\pi)^{d/2} \Upsilon(0), \end{aligned}$$

where we have applied the Plancherel theorem (see (2.11)) and the Fourier transform for Riesz kernel (in the generalized sense):

$$\mathcal{F}(|\cdot|^{-\alpha})(\xi) = \pi^{-d/2} 2^{-\alpha} \frac{\Gamma((d-\alpha)/2)}{\Gamma(\alpha/2)} |\xi|^{-(d-\alpha)}, \quad \text{for } \alpha \in (0, d) \text{ and } \xi \in \mathbb{R}^d.$$

Hence, the second condition in (2.10) can be equivalently written as $L_b^{-2} > (2\pi)^{d/2} \Upsilon(0)$. Comparing this with (1.11b), namely, $L_b^{-2} > 256 \Upsilon(0)$, our condition is sharper when $d > \frac{16 \log(2)}{\log(2\pi)} \approx 6.03$.

Case II: If $\widehat{f}(\cdot)$ is not strictly positive, but only nonnegative, namely, $\widehat{f}(\xi) = 0$ for some $\xi \in \mathbb{R}^d$, how to remove the absolute value in (2.10) becomes tricky, which will be illustrated in the following example:

Example 2.11. Let $d = 1$ and $g(x) = \frac{1}{2} 1_{[-1,1]}(x)$. It is clear that $\widehat{g}(\xi) = \xi^{-1} \sin(\xi)$. Now set $f(x) = (g * g)(x) = 2^{-2} \max(2 - x, 0)$. It is clear that f is nonnegative. It is also nonnegative-definite because $\widehat{f}(\xi) = \widehat{g}(\xi)^2 = \xi^{-2} \sin^2(\xi) \geq 0$. But in this case, $\widehat{f}(\cdot)$ is only nonnegative (not strictly positive) with infinite many zeros. Hence, when taking the square root of $\widehat{f}(\xi)$ as in (2.10), one needs to wisely choose the correct positive and negative branches:

1. Clearly, the signed version $\sqrt{\widehat{f}(\xi)} = \xi^{-1} \sin(\xi)$ is preferable since its inverse Fourier transform can be easily computed, which is equal to $g(x)$. Moreover, because this inverse Fourier transform $g(x)$ is clearly nonnegative, the absolute value signs in (2.10) do not pose any additional restrictions.
2. However, if one chooses the positive branches, namely, $\sqrt{\widehat{f}(\xi)} = |\xi^{-1} \sin(\xi)|$, it is not clear how to compute its Fourier transform. In general, some bad choices of the positive/negative branches may make the conditions in (2.10) fail. For example, such choice may turn $\sqrt{\widehat{f}(\xi)}$ into a distribution, then taking the absolute value of a distribution (unless it is a measure) may be problematic. Or even one may find some choice such that $\sqrt{\widehat{f}(\xi)}$ is a well-defined function, it may well be possible to be signed and after taking absolute values, the integral in (2.10) may blow up.

3 Moment Estimates – Proof of Theorem 1.2

(S:Lrho) We first state some known results and prove a moment bound in Corollary 3.3.

(T:Solve) **Theorem 3.1** (Theorem 1.2 [CH19]). *Suppose that*

- (i) *the initial measure μ is an $\mathcal{F}_0$ -measurable random variable that satisfies the following integrability condition:*

$$\mathbb{E} \left(\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) \right) < \infty \quad \text{for all } a > 0, \quad (3.1) \quad \boxed{\text{A:icon}}$$

(ii) the spectral measure $\hat{f}$ satisfies Dalang's condition (1.12),

Then (1.1) has a unique random field solution starting from μ . Moreover, the solution is $L^2(\Omega)$ continuous and is adapted to the filtration $\{\mathcal{F}_t\}_{t \geq 0}$.

Theorem 3.2 (Theorem 1.7 [CH19]). Under the assumptions of Theorem 3.1, for any $t > 0$, $x \in \mathbb{R}^d$ and $p \geq 2$, the solution to (1.1), $u(t, x)$, given by (1.5) is in $L^p(\Omega)$ and

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(G(t, \cdot) * |\mu|)(x)] H(t; \gamma_p)^{1/2}, \quad (3.2) \quad \boxed{\text{E:mb}}$$

where $\bar{\varsigma} = L_0/L_b$, $\gamma_p = 32pL_b^2$ (see (1.4) for L_0 and L_b) and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the expression of the function H).

Corollary 3.3. Under the same setting as Theorem 3.2, if the two conditions in (1.11) hold (see also (1.15)), then

$$\|u(t, x)\|_p \leq C_p \left(1 + (G(t, \cdot) * |\mu|)(x) \right), \quad \text{for all } p \text{ s.t. } 1/p \in (128L_b^2\Upsilon(0), 1), \quad (3.3) \quad \boxed{\text{E:mb0}}$$

where $C_p = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$.

Proof. Lemma 2.5 of [CK19] gives one sufficient condition, namely, $2\gamma_p\Upsilon(0) < 1$, under which the function $H(t; \gamma_p)$ is bounded in t . Therefore, by taking into account the expression of γ_p in Theorem 3.2, we see that as a direct consequence of (3.2), whenever

$$32pL_b^2 < \frac{1}{4\Upsilon(0)}, \quad (3.4) \quad \boxed{\text{E:pLip?}}$$

we have all p -th moments ($p \geq 2$) bounded as given in (3.3). $\square$

Now we are ready to prove Theorem 1.2.

Proof of Theorem 1.2. Under condition (iii), we can apply Fubini's Theorem and the moment bound (3.3) below to see that for some constant $C > 0$ independent of t , which may vary from line to line,

$$\begin{aligned} \mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) &\leq C \mathbb{E} \left[\int_{\mathbb{R}^d} \left(1 + (G(t, \cdot) * |\mu|)(x) \right)^2 \rho(x) dx \right] \\ &= C \int_{\mathbb{R}^d} \mathbb{E} \left[\left(\left(G(t, \cdot) * (1 + |\mu|) \right)(x) \right)^2 \right] \rho(x) dx \\ &= C \mathcal{G}_\rho(t; \mu) < \infty. \end{aligned}$$

This proves Theorem 1.2. $\square$

Remark 3.4 (Restarted SHE). Recall that the Markov property of the solution to (1.1) implies that for any $t \geq t_0 > 0$,

$$u(t + t_0, x; \mu) \stackrel{\mathcal{L}}{=} u(t, x; u(t_0, \cdot; \mu)) =: v(t, x), \quad (3.5) \quad \boxed{\text{E:eqLaw}}$$

where $\mathcal{L}$ refers to the equality in law and we have used $v(t, x)$ to simplify the notation. It is known that v satisfies the following restarted SPDE:

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}_{t_0}(t, x) & x \in \mathbb{R}^d, t > 0, \\ v(0, x) = u(t_0, x; \mu), & x \in \mathbb{R}^d, \end{cases} \quad (3.6) \quad \boxed{\text{E:restart}}$$

where $\dot{W}_{t_0}(t, x) := \dot{W}(t + t_0, x)$ denotes the time shifted noise, i.e.,

$$\int_0^t \int_{\mathbb{R}^d} W_{t_0}(ds, dy) = \int_{t_0}^{t+t_0} \int_{\mathbb{R}^d} W(ds, dy). \quad (3.7) \quad \boxed{\text{E:ShiftW}}$$

Under conditions in (1.11), Theorem 3.2 and (3.5) imply immediately that

$$\|v(t, x)\|_q = \|u(t + t_0, x; \mu)\|_q \leq C_q \left(1 + (G(t + t_0, \cdot) * |\mu|)(x) \right) = C_q J_0(t + t_0, x; \mu^*), \quad (3.8) \quad \boxed{\text{E:MomBddRst?}}$$

for all $q \geq 2$ and $t > 0$, where the constant C_q does not depend on t . Moreover, under the assumptions of Theorem 1.2, we have $v(0, \cdot) \in L_\rho^2(\mathbb{R}^d)$ a.s. and

$$\mathbb{E} \left(\|v(t, x)\|_\rho^2 \right) = \mathbb{E} \left(\|u(t + t_0, x; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t + t_0; \mu) < \infty. \quad (3.9) \quad \boxed{\text{E:MomRhoBddRst?}}$$

4 A Factorization Lemma

$\boxed{\text{S:Factor?}}$

In this section, we establish a factorization lemma with corresponding moment estimates; see Lemmas 4.2 and 4.4 below. This factorization lemma was first discovered in [DKZ87]; check also Section 5.3.1 of [DZ14]. For $\alpha \in (0, 1)$, $t > 0$ and $x \in \mathbb{R}^d$, define formally

$$\begin{aligned} (F_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{\alpha-1} G(t-s, x-y) f(s, y) ds dy \quad \text{and} \\ (Y_\alpha f)(t, x) &:= \int_0^t \int_{\mathbb{R}^d} (t-s)^{-\alpha} G(t-s, x-y) f(s, y) W(ds, dy). \end{aligned} \quad (4.1) \quad \boxed{\text{E:Ops?}}$$

For F_α , the Step 1 of the proof of [TZ98, Theorem 3.1] showed the following proposition:

$\boxed{\text{P:F-Comp}}$

Proposition 4.1. *Let ρ and $\tilde{\rho}$ be given in condition (i) of Theorem 1.3 (see (1.18)). For any $q > 2$, $t_0 > 0$ and $\alpha \in (q^{-1}, 2^{-1})$, the operator F_α as an operator from $L^q((0, t_0); L_\rho^2(\mathbb{R}^d))$ to $L_\rho^2(\mathbb{R}^d)$ is compact.*

As for Y_α , we have the following two lemmas, which hold for both non-restarted SHE ($t_0 = 0$) and the restarted SHE ($t_0 > 0$):

$\boxed{\text{L:yineq}}$

Lemma 4.2. *Suppose that μ — the initial condition for u — satisfies (3.1) and $\hat{f}$ satisfies Dalang's condition (1.12). Assume that for some $\alpha \in (0, 1/2)$, $\hat{f}$ satisfies the following condition*

$$\mathcal{H}_\alpha(t) := \int_0^t dr r^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-r|\xi|^2) < \infty \quad \text{for all } t > 0. \quad (4.2) \quad \boxed{\text{E:H}}$$

Fix an arbitrary $t_0 \geq 0$. Let $v(t, x)$ be the solution to the restarted SHE (3.6) and $\dot{W}_{t_0}$ be the time-shifted noise (see (3.7)) when $t_0 > 0$ and let $v = u$ when $t_0 = 0$. Then

$$Y_v(s, y) := [Y_\alpha b(\circ, v(\cdot, \circ))](s, y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \quad (4.3) \quad \boxed{\text{E:Y?}}$$

has the following properties:

(1) For all $q \geq 2$, $s > 0$, $y \in \mathbb{R}^d$,

$$\|Y_v(s, y)\|_q^2 \leq H(s + t_0; 32qL_b^2) J_0^2(s + t_0, y; \mu^*) \mathcal{H}_\alpha(s) < \infty, \quad (4.4) \quad \boxed{\text{E:NormY}}$$

where we refer to Theorem 3.2 for the function $H(t; \gamma)$;

- (2) Under two conditions in (1.11), if the integral in (4.2) is finite for some $\alpha \in (128L_b^2\Upsilon(0), 1/2)$, then for any q with $1/q \in (128L_b^2\Upsilon(0), \alpha)$, the function $H(t; 32qL_b^2)$ in (4.4) is uniformly bounded in $t \geq 0$, i.e., $\sup_{t \geq 0} H(t; 32qL_b^2) < \infty$;
- (3) Under two conditions in (1.11), if the integral in (4.2) is finite for some $\alpha \in (128L_b^2\Upsilon(0), 1/2)$, then for any q with $1/q \in (128L_b^2\Upsilon(0), \alpha)$ and for any nonnegative and $L^1(\mathbb{R}^d)$ -function ρ , there exists a constant $\Theta = \Theta(q, L_b, L_0, \alpha)$, which does not depend on t , such that for $t > 0$,

$$\mathbb{E} \left(\int_0^t \|Y_v(s, \cdot)\|_\rho^q ds \right) \leq \Theta \int_0^t [\mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s)]^{q/2} ds, \quad (4.5) \quad \boxed{\text{E:yineq}}$$

which is finite provided that

$$\int_0^t [\mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s)]^{q/2} ds < \infty. \quad (4.6) \quad \boxed{\text{E:GH-int'}}$$

Remark 4.3. Condition (4.6) is true for $t_0 > 0$ because $\mathcal{G}_\rho(t; \mu)$ is a continuous function for $t > 0$ and $\mathcal{H}_\alpha(s)$ is continuous and bounded for $s \in [0, t]$ thanks to (4.2). However, when $t_0 = 0$, the situation is much more trickier. For example, when the initial condition is the delta initial condition, we have that

$$\mathcal{G}_{\tilde{\rho}}(t; \delta_0) = \int_{\mathbb{R}^d} G(t, x)^2 \rho(x) dx = G(2t, 0) \int_{\mathbb{R}^d} G(t/2, x) \rho(x) dx < \infty,$$

where one can obtain the second equality via (1.21). Hence, when $s \rightarrow 0$, $\mathcal{G}_{\tilde{\rho}}(s; \delta_0)$ blows up with a rate $s^{-d/2}$. On the other hand, $\mathcal{H}_\alpha(s)$ goes to zero with a different rate. One needs to combine these two rates to check if condition (4.6) holds. By introducing t_0 (or restart the heat equation), one can avoid this issue (of the potential singularity of $\mathcal{G}_{\tilde{\rho}}$ at $s = 0$).

Proof. In the proof, we use C to denote a generic constant that may change its value at each appearance.

(1) We first prove (4.4). By the Burkholder-Davis-Gundy inequality and Minkowski's integral inequality, we see that

$$\begin{aligned} \|Y_v(s, y)\|_q^2 &\leq C \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(z_1, v(r, z_1))\|_q \\ &\quad \times f(z_1 - z_2) G(s-r, y-z_2) \|b(z_2, v(r, z_2))\|_q. \end{aligned}$$

Note that for the Lipschitz condition in (1.4), we have that

$$|b(x, v)| \leq |b(x, v) - b(x, 0)| + |b(x, 0)| \leq L_b |v| + L_0 \leq C(1 + |v|), \quad C := L_b \vee L_0.$$

We apply this and the moment bound (3.2) to $\|b(z_i, v(r, z_i))\|_q$ above to see that

$$\begin{aligned} \|b(z_i, v(r, z_i))\|_q &\leq C \left(1 + \|v(r, z_i)\|_q \right) \\ &= C \left(1 + \|u(r + t_0, z_i)\|_q \right) \\ &\leq CH(r + t_0; 32qL_b^2) J_0(r + t_0, z_i; \mu^*) \\ &\leq CH(s + t_0; 32qL_b^2) J_0(r + t_0, z_i; \mu^*), \quad i = 1, 2, r \in (0, s), \end{aligned} \quad (4.7) \quad \boxed{\text{E:NormB}}$$

where in the last step, we have used the fact that $H(t; \gamma)$ is a nondecreasing function; see Lemma 2.6 of [CK19]. Therefore, by denoting $C_s := H(s + t_0; 32qL_b^2)$,

$$\begin{aligned}
\|Y_v(s, y)\|_q^2 &\leq CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) J_0(r+t_0, z_i; \mu^*) \right) \\
&= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \prod_{i=1}^2 \left(G(s-r, y-z_i) G(r+t_0, z_i - \sigma_i) \right) \\
&= CC_s \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s+t_0, y - \sigma_1) G(s+t_0, y - \sigma_2) \\
&\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{(r+t_0)(s-r)}{s+t_0}, z_i - \sigma_i \frac{r+t_0}{s+t_0} - \frac{s-r}{s+t_0} y\right) \\
&\leq CC_s (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} \mu^*(d\sigma_1) \mu^*(d\sigma_2) G(s+t_0, y - \sigma_1) G(s+t_0, y - \sigma_2) \\
&\quad \times \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right),
\end{aligned}$$

where we have applied (1.21) and Plancherel's theorem. Hence,

$$\|Y_v(s, y)\|_q^2 \leq CC_s (2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{(r+t_0)(s-r)}{s+t_0} |\xi|^2\right).$$

Because the function

$$t_0 \mapsto \frac{r+t_0}{s+t_0} = 1 - \frac{s-r}{s+t_0} \quad \text{for } t_0 > 0,$$

is nondecreasing in t_0 whenever $s > r > 0$, we can replace the two appearances of t_0 in the exponent of the above inequality by zero to see that

$$\|Y_v(s, y)\|_q^2 \leq CC_s (2\pi)^{-2d} J_0^2(s+t_0, y; \mu^*) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \quad (4.8) \quad \boxed{\text{E_NormY}}$$

Furthermore, by symmetry of $r(s-r)/s$ and the fact that $r(s-r)/s \geq r/2$ for all $r \in [0, s/2]$, we see that the above double integral is bounded by

$$\begin{aligned}
&\leq 2 \int_0^{s/2} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r}{2} |\xi|^2\right) \\
&= 2^{2(1-\alpha)} \int_0^{s/4} dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\
&\leq 2^{2(1-\alpha)} \int_0^s dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r |\xi|^2) \\
&= 2^{2(1-\alpha)} \mathcal{H}_\alpha(s).
\end{aligned}$$

Plugging the above bound back to (4.8) proves (4.4).

(2–3) Part (2) is a direct consequence of Theorem 3.2. It remains to prove (4.5). An application of Minkowski's inequality shows that

$$\mathbb{E} \left(\|Y_v(s, \cdot)\|_\rho^q \right) = \left\| \int_{\mathbb{R}^d} Y_v(s, y)^2 \rho(y) dy \right\|_{q/2}^{q/2} \leq \left(\int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \right)^{q/2}. \quad (4.9) \quad \boxed{\text{E_EYqr}}$$

By the definition of $\mathcal{G}_\rho(t; \mu)$ in (1.16) and by (4.4), we see that

$$\int_{\mathbb{R}^d} \|Y_v(s, y)\|_q^2 \rho(y) dy \leq C \mathcal{G}_\rho(s + t_0; \mu) \mathcal{H}_\alpha(s).$$

Plugging the above expression to the far right-hand of (4.9) proves (4.5). Finally, the finiteness of the upper bound in (4.5) is guaranteed by condition (4.6). This completes the proof of Lemma 4.2. $\square$

(L:Factorize)

Lemma 4.4 (Factorization lemma). *Suppose that μ — the initial condition for u — satisfies (3.1) and $\hat{f}$ satisfies Dalang's condition (1.12). Assume that condition (4.2) is satisfied for some $\alpha \in (0, 1/2)$. Fix an arbitrary $t_0 \geq 0$. Let $v(t, x)$ be the solution to the restarted SHE (3.6) and $\tilde{W}_{t_0}$ be the time-shifted noise (see (3.7)) when $t_0 > 0$ and let $v = u$ when $t_0 = 0$. Then the following factorization holds*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz),$$

for all $t > 0$ and $x \in \mathbb{R}^d$. As a consequence,

$$v(t, x) = [G(t, \cdot) * u(t_0, \cdot; \mu)](x) + \frac{\sin(\alpha\pi)}{\pi} [F_\alpha Y_v](t, x), \quad \text{for all } t > 0 \text{ and } x \in \mathbb{R}^d. \quad (4.10) \text{ ?E:1fac?}$$

Proof. The lemma is straightforward provided that one can switch the orders of stochastic and ordinary integrals:

$$\begin{aligned} & \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y_v(s, \cdot)](x) ds \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \\ & \quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(z, v(r, z)) W_{t_0}(dr, dz) \\ &= \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz) \end{aligned} \quad (4.11) \text{ [E:Fubini_1]}$$

$$\begin{aligned} &= \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(z, v(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &= \frac{\pi}{\sin(\alpha\pi)} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(z, v(r, z)) W_{t_0}(dr, dz), \end{aligned} \quad (4.12) \text{ [E:Fubini_2]}$$

where the last step is the *Beta integral* which requires that $\alpha \in (0, 1)$. It remains to justify the two applications of the stochastic Fubini's theorem (see Theorem 5.30 of Chapter one in [Dal+09], or also [Wal86] or Theorem 4.33 of [DZ14]) in (4.11) and (4.12) in the following two steps.

Step 1. In this step, we justify the change of orders in (4.11). Note that t, x and s are fixed. It suffices to prove the following condition:

$$\begin{aligned} I_1 &:= \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\ & \quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(s-r, y-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, v(r, z_i)) \right) \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y_v(s, y)\|_2^2 \\
&< +\infty.
\end{aligned}$$

But this follows immediately from (4.4). Indeed

$$\begin{aligned}
\int_{\mathbb{R}^d} dy G(t-s, x-y) \|Y_v(s, y)\|_2^2 &\leq C \int_{\mathbb{R}^d} dy G(t-s, x-y) J_0^2(s+t_0, y; \mu^*) \mathcal{H}_\alpha(s) \\
&= C \mathcal{H}_\alpha(s) \int_{\mathbb{R}^d} dy G(t-s, x-y) \iint_{\mathbb{R}^{2d}} \mu^*(dz_1) \mu^*(dz_2) \\
&\quad \times G(s+t_0, y-z_1) G(s+t_0, y-z_2).
\end{aligned}$$

Now we bound the three heat kernels using (1.21) as follows:

$$\begin{aligned}
&G(t-s, x-y) \prod_{i=1}^2 G(s+t_0, y-z_i) \\
&= \frac{G(2(t-s), x-y)^2}{G(4(t-s), 0)} \prod_{i=1}^2 G(s+t_0, y-z_i) \\
&\leq 2^d \frac{G(2(t-s), x-y)^2}{G(4(t-s), 0)} \prod_{i=1}^2 G(2s+2t_0, y-z_i) \\
&= 2^d [4(t-s)]^{d/2} \prod_{i=1}^2 \left[G(2s+2t_0, y-z_i) G(2(t-s), x-y) \right] \\
&= 2^{2d} (t-s)^{d/2} \prod_{i=1}^2 \left[G(2(t+t_0), x-z_i) G\left(\frac{2(t-s)(s+t_0)}{t+t_0}, y - \frac{s+t_0}{t+t_0}(x-z_i)\right) \right] \\
&\leq 2^{2d} (t-s)^{d/2} \prod_{i=1}^2 \left[G(2(t+t_0), x-z_i) G\left(\frac{2(t-s)(s+t_0)}{t+t_0}, 0\right) \right] \\
&\leq C_{t,s,t_0} \prod_{i=1}^2 G(2(t+t_0), x-z_i).
\end{aligned}$$

Therefore, $I_1 \leq C_{t,s,t_0} \mathcal{H}_\alpha(s) J_0^2(2(t+t_0), x; \mu^*) < \infty$.

Step 2. Similarly, as for (4.12), we need to show that

$$\begin{aligned}
I_2 &:= \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 \\
&\quad \times f(z_1 - z_2) \left(\prod_{i=1}^2 G(t-r, x-z_i) \right) \mathbb{E} \left(\prod_{i=1}^2 b(z_i, v(r, z_i)) \right) < \infty.
\end{aligned}$$

By the Cauchy Schwartz inequality, (4.7) and because $\alpha \in (0, 1/2)$,

$$\begin{aligned}
I_2 &\leq C \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x-z_i) J_0(r+t_0, z_i; \mu^*) \right) \\
&= C' \int_0^t dr (t-r)^{-\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 \left(G(t-r, x-z_i) J_0(r+t_0, z_i; \mu^*) \right).
\end{aligned}$$

Now by the same arguments as those leading to (4.4) (with 2α there replaced by α), we see that

$$I_2 \leq C J_0^2(t, x; \mu^*) \int_0^t dr r^{-\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right),$$

which is finite by (4.2) where we replace α with $\alpha/2$ and repeat the same steps right after (4.8). This completes the proof of Lemma 4.4. $\square$

Finally, we characterize conditions (1.14) and (4.2) in the following lemma:

$\langle \text{L:HUA} \rangle$ **Lemma 4.5.** *For all $\alpha \in (0, 1/2]$, we have the following properties:*

(1) $(2\pi)^{-d} \mathcal{H}_\alpha(t) \leq \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$ for all $t > 0$ and hence

$$\Upsilon_{2\alpha}(0) < \infty \implies \mathcal{H}_\alpha(t) < \infty \text{ for all } t > 0; \quad (4.13) \quad \boxed{\text{E:HUA}}$$

(2) $\lim_{t \rightarrow \infty} (2\pi)^{-d} \mathcal{H}_\alpha(t) = \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$;

(3) if $\Upsilon(0) < \infty$, then the reverse implication of (4.13) holds.

Proof. We only need to consider the case when $\alpha > 0$. It is clear that the function $\mathcal{H}_\alpha(t)$ is nondecreasing. Hence, (2) implies (1). As for (2), by Fubini's theorem, we see that

$$\lim_{t \rightarrow \infty} \mathcal{H}_\alpha(t) = \int_0^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) e^{-r|\xi|^2} = \Gamma(1 - 2\alpha) \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^{2(1-2\alpha)}} = C \Upsilon_{2\alpha}(0), \quad (4.14) \quad \boxed{\text{E:Fubini}}$$

with $C := \Gamma(1 - 2\alpha)(2\pi)^d$. As for (3), from (4.14), for any $t > 0$, by splitting the dr integral into two parts we see that

$$\begin{aligned} C \Upsilon_{2\alpha}(0) &= \int_0^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) = \mathcal{H}_\alpha(t) + I_\alpha(t), \quad \text{with} \\ I_\alpha(t) &= \int_t^\infty dr r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2). \end{aligned}$$

Notice that

$$I_\alpha(t) \leq t^{-2\alpha} \int_t^\infty dr \int_{\mathbb{R}^d} \widehat{f}(d\xi) e^{-r|\xi|^2} = t^{-2\alpha} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^2} e^{-t|\xi|^2} \leq t^{-2\alpha} \int_{\mathbb{R}^d} \frac{\widehat{f}(d\xi)}{|\xi|^2} = \frac{(2\pi)^d}{t^{2\alpha}} \Upsilon(0).$$

Therefore,

$$\Upsilon_{2\alpha}(0) \leq \frac{\mathcal{H}_\alpha(t)}{(2\pi)^d \Gamma(1 - 2\alpha)} + \frac{\Upsilon(0)}{\Gamma(1 - 2\alpha) t^{2\alpha}} < \infty, \quad \text{for all } t > 0,$$

which proves (3). $\square$

5 Tightness and Construction – Proof of Theorem 1.3

$\langle \text{S:mainproof} \rangle$ 5.1 Proof of part (a) of Theorem 1.3

We are now ready to prove part (a) of Theorem 1.3.

Proof of Theorem 1.3 (a). In this proof, $u(t, x)$ refers to $u(t, x; \mu)$. Fix $\tau > 0$ and $t_0 = \tau/2$. Throughout the proof, we have $t \geq \tau$. Let v be the solution to (3.6) that is restarted from $t - t_0$. Then

$$v(t_0, \cdot) \stackrel{\mathcal{L}}{=} u(t, \cdot) \quad \text{for } t \geq \tau.$$

According to Assumption (i), we can choose and fix some admissible weight function $\tilde{\rho}$ such that (1.18) is satisfied. Hence, by Proposition 2.2, the following set

$$\mathcal{K}_1(\Lambda) := \left\{ (G(t_0, \cdot) * y(\cdot))(x) : \|y\|_{\tilde{\rho}} \leq \Lambda \right\} \quad \text{with } \Lambda > 0$$

is relative compact in $L^2_{\tilde{\rho}}(\mathbb{R}^d)$.

Assumption (iii), i.e., (1.11), implies that the interval $(2^7 L_b^2 \Upsilon(0), 1/2)$ is not empty. Moreover, Assumption (iv), i.e., (1.14), guarantees that there exists a constant α in this interval, namely, $2^7 L_b^2 \Upsilon(0) < \alpha < 1/2$ such that (1.14) holds with α replaced by 2α , i.e., $\Upsilon_{2\alpha}(0) < \infty$. Now we can apply Part (3) of Lemma 4.5, thanks to (1.11a), to see that (4.2) holds. Therefore, both Lemmas 4.4 and 4.2 (in particular, the part (3)) are applicable. In particular, Lemma 4.4 ensures that the following factorization is well defined:

$$v(t_0, x) = \left(G(t_0, \cdot) * u(t - t_0, \cdot) \right)(x) + \frac{\sin(\alpha\pi)}{\pi} [F_{\alpha} Y_v](t_0, x). \quad (5.1) \quad \boxed{\text{E:Fact}}$$

Part (3) of Lemma 4.2 shows that for any q in the following range:

$$128 L_b^2 \Upsilon(0) < \frac{1}{q} < \alpha < \frac{1}{2} \quad \left(\text{or equivalently } 2 < \frac{1}{\alpha} < q < \frac{1}{128 L_b^2 \Upsilon(0)} \right), \quad (5.2) \quad \boxed{\text{E:Choice}}$$

we can apply Proposition 4.1 to see that the following set

$$\mathcal{K}_2(\Lambda) := \left\{ (F_{\alpha} h)(x) : \|h\|_{L^q((0, t_0); L^2_{\tilde{\rho}}(\mathbb{R}^d))} \leq \Lambda \right\} \quad \text{with } \Lambda > 0$$

is relative compact in $L^2_{\tilde{\rho}}(\mathbb{R}^d)$. Now for any $\Lambda > 0$, define the set $\mathcal{K}(\Lambda)$ as

$$\begin{aligned} \mathcal{K}(\Lambda) &:= \mathcal{K}_1(\Lambda) + \mathcal{K}_2(\Lambda) \\ &= \left\{ (G(t_0, \cdot) * y(\cdot))(x) + (F_{\alpha} h)(x) : \|y\|_{\tilde{\rho}} \leq \Lambda \quad \text{and} \quad \|h\|_{L^q((0, t_0); L^2_{\tilde{\rho}}(\mathbb{R}^d))} \leq \Lambda \right\}. \end{aligned}$$

Notice that from the factorization formula (5.1),

$$\begin{aligned} \mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] &\leq \mathbb{P} \left[\left(\int_0^{t_0} \|Y_v(s, \cdot)\|_{\tilde{\rho}}^q ds \right)^{1/q} > \frac{\pi \Lambda}{\sin(\alpha\pi)} \right] + \mathbb{P} \left[\|u(t - t_0, \cdot)\|_{\tilde{\rho}} > \Lambda \right] \\ &=: I_1 + I_2. \end{aligned}$$

By Chebyshev's inequality and (1.17), we see that

$$I_2 \leq \frac{1}{\Lambda^2} \mathbb{E} \left(\|u(t - t_0, \cdot)\|_{\tilde{\rho}}^2 \right) \leq \frac{1}{\Lambda^2} \mathcal{G}_{\tilde{\rho}}(t - t_0; \mu).$$

Because $\mathcal{G}_{\tilde{\rho}}(t; \mu)$ is a continuous function for $t > 0$, and because it is also bounded at infinity thanks to Assumption (ii) (see (1.19)), we have that

$$\mathcal{G}_{\tilde{\rho}}(t - t_0; \mu) \leq \sup_{t \geq \tau} \mathcal{G}_{\tilde{\rho}}(t - t_0; \mu) = \sup_{t \geq t_0} \mathcal{G}_{\tilde{\rho}}(t; \mu) < \infty. \quad (5.3) \quad \boxed{\text{E:GtBdd}}$$

Therefore, we can bound I_2 from above with a constant that does not depend on $t \geq \tau$, namely,

$$I_2 \leq \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) < \infty.$$

As for I_1 , with the choice of α and q in (5.2), one can apply Chebyshev's inequality and part (3) of Lemma 4.2 to see that

$$I_1 \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \mathbb{E} \int_0^{t_0} \|Y_v(s, \cdot)\|_{\bar{\rho}}^q ds \leq \frac{\sin^q(\alpha\pi)}{\pi^q \Lambda^q} \Theta \int_0^{t_0} (\mathcal{G}_{\bar{\rho}}(s + t - t_0; \mu) \mathcal{H}_{\alpha}(s))^{q/2} ds,$$

where the constant Θ does not depend on t . As we have seen from above that $\Upsilon_{2\alpha}(0) < \infty$, we can apply Lemma 4.5 to bound $\mathcal{H}_{\alpha}(s)$ from above by the finite bound: $(2\pi)^d \Gamma(1 - 2\alpha) \Upsilon_{2\alpha}(0)$. Hence, together with (5.3), we obtain the following upper bound for I_1 that is uniform in $t \geq \tau$:

$$I_1 \leq \frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0).$$

Combining these two upper bounds, we see that

$$\mathbb{P}[v(t_0, \cdot) \notin \mathcal{K}(\Lambda)] \leq \frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0) + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) < \infty,$$

with the upper bound holding uniformly for all $t \geq \tau$. Hence, for any $\epsilon > 0$, by choosing $\Lambda > 0$ big enough such that

$$\frac{\sin^q(\alpha\pi) \Theta (2\pi)^{dq/2} t_0}{\Gamma(1 - 2\alpha)^{q/2} \pi^q \Lambda^q} \left(\sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) \right)^{q/2} \Upsilon_{2\alpha}^{q/2}(0) + \frac{1}{\Lambda^2} \sup_{t \geq t_0} \mathcal{G}_{\bar{\rho}}(t; \mu) < \epsilon,$$

we can ensure that

$$\mathbb{P}(u(t, \cdot) \in \mathcal{K}(\Lambda)) = \mathbb{P}(v(t_0, \cdot) \in \mathcal{K}(\Lambda)) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau,$$

which proves part (a) of Theorem 1.3. $\square$

5.2 Proof of part (b) of Theorem 1.3

The proof of part (b) based on (a) is routine.

Proof of Theorem 1.3 (b). Fix an arbitrary $\tau > 0$ and denote

$$U(T) := \frac{1}{T} \int_{\tau}^{T+\tau} \mathcal{L}(u(t, \cdot; \mu)) dt, \quad T > 0.$$

We claim that the family of laws $U(T, \cdot)$ for $T > 0$ is tight in $L_{\rho}^2(\mathbb{R}^d)$. Indeed, for any $\epsilon \in (0, 1)$, by part (a), there exists a compact set $\mathcal{K} \in L_{\rho}^2(\mathbb{R}^d)$ such that (1.20) holds. This implies that

$$U(T)(\mathcal{K}) = \frac{1}{T} \int_{\tau}^{T+\tau} \mathcal{L}(u(t, \cdot; \mu))(\mathcal{K}) dt \geq \frac{1}{T} \int_{\tau}^{T+\tau} (1 - \epsilon) dt = 1 - \epsilon, \quad \text{for all } T > 0.$$

Let $\{T_n\}_{n \in \mathbb{N}}$ be any deterministic sequence such that $T_n \uparrow \infty$. Since $\{U(T_n)\}_{n \geq 1}$ is a tight sequence of measures, there exists a subsequence $\{U(T_{n_m})\}_{m \geq 1}$ that converges weakly to a measure, say, η , on $L_{\rho}^2(\mathbb{R}^d)$. Then one can apply for the Krylov-Bogoliubov existence theorem (see, e.g., [DZ14, Theorem 11.7]) to conclude that the measure η is the invariant measure for $\mathcal{L}(u(t, \cdot; \mu))$, $t \geq \tau$. Finally, since τ can be arbitrarily close to zero, one can conclude part (b) of Theorem 1.3. $\square$

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